Jiaqi Kevin Yang

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EDUCATION

MSc in Computer Science at McGill University **Sep 2026 - Dec 2027**

- Teaching Assistant for **200+** students in computer architecture, providing targeted low-level code reviews

BSc in Software Engineering at McGill University **Sep 2022 - May 2026**

- Cumulative GPA: 3.74/4.00

EXPERIENCE

QA Automation Intern at dormakaba **May 2025 - Aug 2025**

- Built a custom Locust load testing framework that simulated up to **1000** concurrent users and tested how the backend API behaved under different stress patterns to uncover bottlenecks and scalability issues
- Eliminated over **20** hours of manual regression and integration testing per release cycle by extending a RESTful Flask API on a Raspberry Pi-powered hardware testing jig
- Removed hardcoded credentials across 3 repos by integrating AWS SSO and Parameter Store
- Automated over **50** Java end-to-end validation tests into CI/CD pipelines across firmware and cloud releases

PROJECTS

C Compiler **Jan 2026 - May 2026**

- Built an end-to-end C to MIPS assembly compiler in Java, refactoring C grammar to eliminate left recursion and developing a recursive-descent parser to generate abstract syntax trees
- Engineered multi-pass semantic analysis engine to enforce strong typing, nested scoping and struct alignment alongside a code generator managing calling conventions, stack frame layout and jump offsets
- Implemented Chaitin's graph coloring algorithm to allocate registers with heuristic register spilling to memory

Smart Dumbbell **Nov 2025**

- Engineered a non-blocking, multithreaded Flask backend to process concurrent MQTT hardware streams and a computer vision pipeline to accurately detect and classify different workout exercises
- Designed a JS/HTML/CSS site that polls REST endpoints backed by a SQLite3 database to show live stats

Distributed Middleware System **Sep 2025 - Oct 2025**

- Engineered a distributed system in Java with a middleware server to route concurrent client requests to replicated multi-threaded backend resource managers through a simulated **7** machine distributed network
- Developed a non-blocking TCP socket protocol, enabling async messages to maximize request throughput without thread starvation, alongside atomic remote procedure calls to guarantee **zero** data races

AI Study Assistant **Nov 2024**

- Deployed a local Qwen 2.5 pipeline, integrating a large language model into a full-stack Django web app to generate various study tools such as summaries, flashcards, study podcasts and quizzes
- Integrated OpenAI Whisper, enabling real-time audio transcriptions for the model to process lecture recordings

SKILLS

Programming Languages: Java, Python, C#, JavaScript/TypeScript, HTML, CSS, C, C++, SQL, GraphQL, Bash
Technical Skills: Git, Agile, Linux/Unix, Docker, Cursor, REST APIs, AWS, Kubernetes, PyTorch, Django, Flask
CS Fundamentals: Operating Systems, Distributed Systems, Concurrent Programming, Machine Learning
Languages: Fluent English, Fluent French, Advanced Mandarin Chinese, Beginner Japanese